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**ASSIGNMENT : DIGITAL EVIDENCE HANDLING & CRIME
SCENE PROCESSING**

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PART A: DIGITAL EVIDENCE PRIORITIZATION AND JUSTIFICATION

1. IDENTIFICATION OF POTENTIAL SOURCES OF DIGITAL EVIDENCE

Based on the provided dataset, multiple sources of digital evidence were identified and categorised into volatile (live) and non volatile (static) evidence.

Volatile (live) evidence

Volatile evidence is highly time sensitive and must be prioritized as it can be lost if the system state changes. The presence of these artifacts indicates that the system was actively being used during the accident, potentially for data transfer or exfiltration.

- Active network activity

Evidence : insider_traffic.pcap

- Random Access Memory (RAM)

Contains active processes, encryption keys and session data

- Running processes and user sessions

Volatile evidence also includes RAM contents, active processes and network sessions which are critical as they are lost upon system shutdown.

Non-Volatile (static) evidence

Non-volatile evidence persists after system shutdown and is critical for reconstructing events.

- Confidential data files

Evidence : client_database.csv , salary_records.csv

- Deleted file

Evidence : client_database_deleted.csv

- System logs

Evidence : system_log.txt

- USB activity logs

Evidence : usb_logs.txt

- Browser history

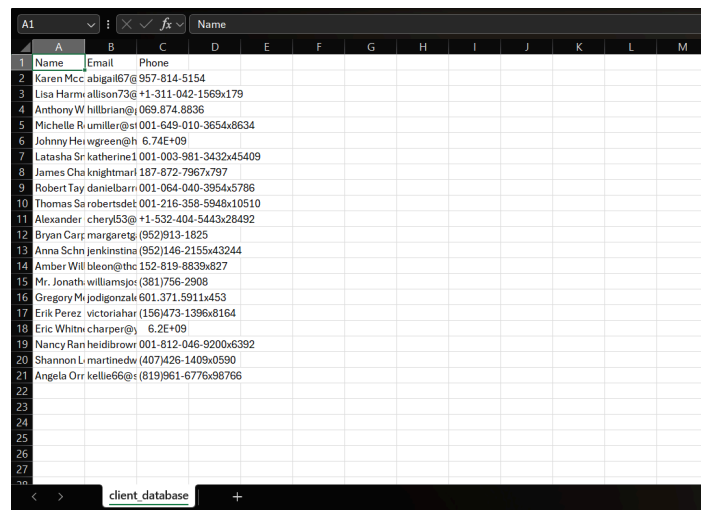
Evidence : history.txt

- Email communication

Evidence : suspicious_email.eml

Although USB devices store non-volatile data, they are considered high risk due to their removable nature and potential for tampering or removal.

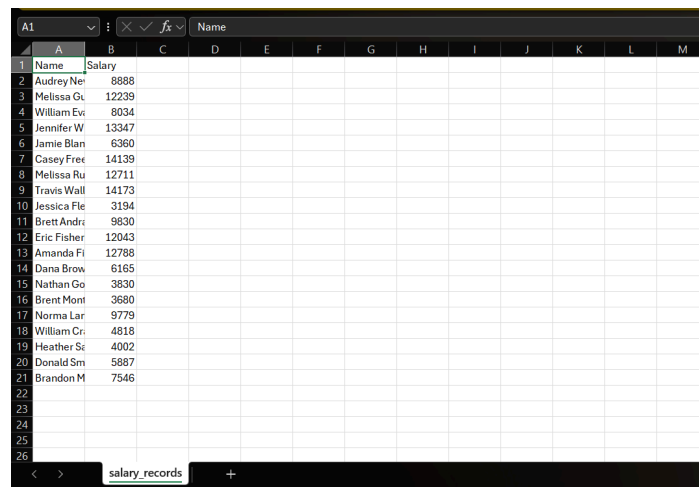
The confidential files contain sensitive organizational data making them likely targets.



A screenshot of a spreadsheet application showing a table with three columns: Name, Email, and Phone. The table contains 21 rows of data, including names like Karen Mcc, Lisa Harms, and Anthony W hillbrian, along with their email addresses and phone numbers. The spreadsheet is titled 'client_database.csv' in the bottom tab.

Name	Email	Phone
Karen Mcc	abigail67@	957-814-5154
Lisa Harms	allison73@	+1-311-042-1569x179
Anthony W hillbrian	069.874.8836	
Michelle R umiller	st001-649-010-3654x8634	
Johnny Hei	wgreen@h	6.74E+09
Latasha Sr	katherine1	001-003-981-3432x45409
James Cha	knightmari	187-872-7967x797
Robert Tay	danielbarr	001-064-040-3954x5786
Thomas Sa	robertsdet	001-216-358-5948x10510
Alexander	chery53@	+1-532-404-5443x28492
Bryan Carj	margaretg	(952)913-1825
Anna Schn	jenkstina	(952)146-2155x43244
Amber Wil	bleon@thc	152-819-8839x827
Mr. Jonath	williamsjor	(381)756-2908
Gregory Mj	jodigonzak	601.371.5911x453
Erik Perez	victoriahar	(156)473-1396x8164
Eric Whitn	charper@y	6.2E+09
Nancy Ran	heidlbrow	001-812-046-9200x6392
Shannon L	martinedw	(407)426-1409x0590
Angela Orr	kellie66@	(819)961-6776x98766

Figure 1 : client_database.csv



A screenshot of a spreadsheet application showing a table with two columns: Name and Salary. The table contains 21 rows of data, including names like Audrey Ner, Melissa Gc, and William Evi, along with their corresponding salary values. The spreadsheet is titled 'salary_records.csv' in the bottom tab.

Name	Salary
Audrey Ner	8888
Melissa Gc	12239
William Evi	8034
Jennifer W	13347
Jamie Blan	6360
Casey Frex	14139
Melissa Ru	12711
Travis Wall	14173
Jessica Fle	3194
Brett Andri	9630
Eric Fisher	12043
Amanda Fi	12788
Dana Brow	6165
Nathan Go	3830
Brent Mont	3680
Norma Lar	9779
William Cr	4818
Heather Sc	4002
Donald Sm	5887
Brandon M	7546

Figure 2 : salary_records.csv

2. EVIDENCE PRIORITIZATION BASED ON VOLATILITY

The prioritization follows the Order of Volatility principle as recommended by NIST guidelines, which emphasizes collecting the most volatile data first to prevent loss of critical evidence.

1. RAM and active system memory
2. Network traffic (insider_traffic.pcap)
3. Running processes and user sessions
4. USB activity (usb_log.txt)
5. System logs (system_log.txt)
6. Deleted data (client_database_deleted.csv)
7. Persistence files (client_database.csv , salary_records.csv , history.txt , suspicious_email.eml)

3. JUSTIFICATION OF PRIORITIZATION

The prioritization is based on the risk of data loss and relevance to the insider attack scenario:

- Volatile evidence such as RAM and active network connections are prioritized first due to their high risk of loss. If the system is powered off, this data will be permanently destroyed.
- USB activity is critical, as shown in Figure 3, where a SanDisk USB device was inserted at a specific timestamp, indicating possible data transfer.

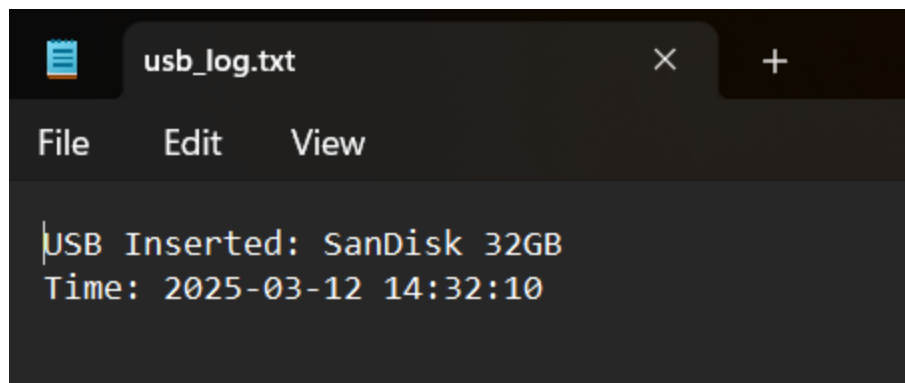
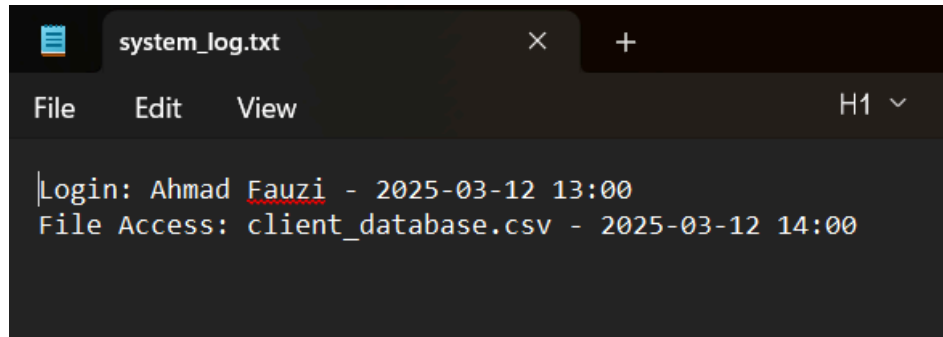


Figure 3 : USB log showing external device insertion (SanDisk 32GB) with timestamp

- System logs, shown in Figure 4, confirms that Ahmad Fauzi logged into the system and accessed client_database.csv. This indicates the starting point of the incident and establishes user involvement.



- Deleted file evidence, as shown in Figure 5, indicates that data was intentionally removed.

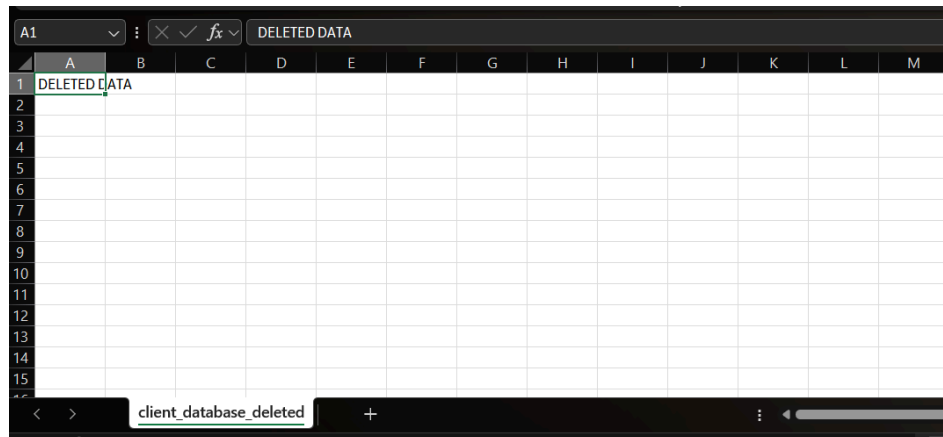


Figure 5 : Deleted file indicating removal of original confidential data

- Email evidence, shown in Figure 6 is critical in establishing intent. The instruction “Delete after use” demonstrates awareness of wrongdoing and deliberate effort to conceal evidence.

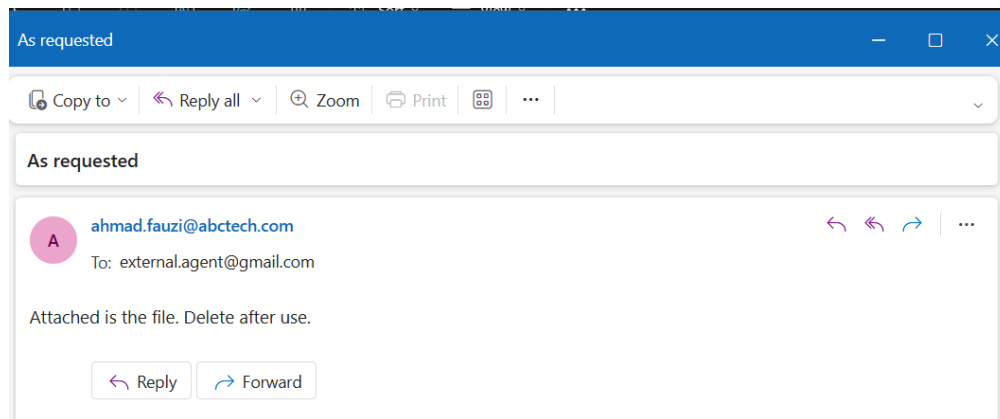


Figure 6 : Email showing file transfer to external party with instruction “Delete after use.”

- Browser history, shown in Figure 7, reveals access to cloud storage platforms suggesting possible alternative methods of data exfiltration.

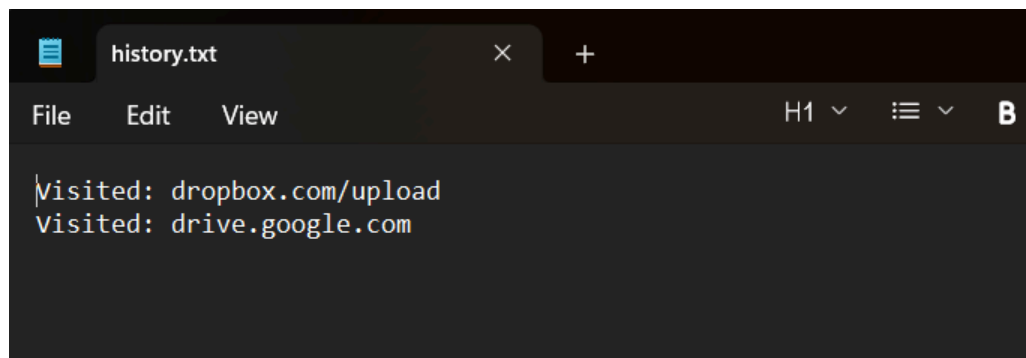


Figure 7 : Browser history showing access to Dropbox and Google Drive upload pages

PART B : REAL-TIME CRIME SCENE RESPONSE STRATEGY

1. Real-Time Response Upon Arrival

My first thought on arriving at the suspected compromised workstation is to safeguard and preserve digital evidence without changing the system state. I start with a visual and situational evaluation of the environment. This involves monitoring whether the system is turned on, whether there are any active applications on the screen, and whether the system has any connected devices like USB devices, external storage, or mobile devices.

At this point, I do not engage with the system to avoid any evidence being changed accidentally. Rather, I capture the whole situation by taking photos of the workstation, screen, cable connections, and the immediate surrounding environment. I also note the time in which the system is in operation and time discrepancies are deemed critical in a forensics investigation.

At the same time, I recognize people who are at the scene and make sure that no one is touching the device. This is an initial measure to make sure that I comprehend the state of the system before proceeding any further.

2. Decision : Live Acquisition or Immediate Shutdown

The option I would take in this case is conducting live acquisition as opposed to just closing the system. It is also assumed that the system is already turned on, meaning that valuable volatile data that is stored in RAM is available at the moment. This includes:

- Running processes
- Active network connections
- Logged-in user sessions
- Encryption keys
- Any possible malware in memory

If I immediately shut down the system, all volatile information would be lost forever and possibly all of the volatile information regarding the unauthorized access or data exfiltration.

Live acquisition will help me to capture this important data before it is lost, and it is of particular importance in the cases that involve:

- Insider threats
- Unauthorized access
- Malware infection

- Ongoing network communication

But I also recognize that in some cases, immediate shutdown can be reasonable, especially when there are threats of active data destruction, such as ransomware or a self-destructing script. Here, the most suitable and forensically sound decision is live acquisition since there is no immediate sign of such threats.

3. Execution of Live Acquisition

Having made the decision regarding live acquisition, I will go about it with the help of trusted forensic tools that will be saved on outside media to reduce the amount of contamination of the system. I start by recording:

- To save volatile data, a memory dump (RAM acquisition).
- An overview of ongoing processes and services in operation.
- Existing connections to the network and open ports.

All the tools involved are tested and proven to be forensically accepted such that the process does not affect system data significantly. All the steps performed in the given stage are well noted along with the tools employed and the commands that were run.

4. Controlling the Environment

I have stringent limitations on individuals and equipment to ensure that I have control of the crime scene.

1. People Control

I will immediately quarantine the workstation area and make sure that only authorized persons are in the area. Any people who are in close proximity to the system are requested to move away to avoid disturbance. I also note down who might be a witness and record their attendance to be further examined.

2. Device Control

I disconnect the system with the external communication by unplugging Ethernet cable or turning off Wi-Fi. This is important because it is necessary to:

Stop hackers accessing the computer remotely.

Prevent continued data exfiltration.

Remote evidence tampering should be avoided.

Importantly, I do not turn off the system because it would make no sense to decide to retain volatile data.

5. Preventing Evidence Contamination

To make sure that evidence is reliable I follow several precautions that will help to prevent contamination:

- I do not needlessly interact with the system, which may be opening files or running irrelevant commands.
- Acquisition is based on the use of forensically sound tools.
- I make sure that the external storage devices to be used are clean and in readiness.
- Physical handling of devices is minimized and done carefully

I also keep the footprint approach at a minimal level, like I take only the necessary steps that are necessary to gather the evidence, and this way, I reduce the risk of interfering with original data.

6. Maintaining Forensic Integrity

Forensic integrity is critical in order to ensure that the evidence can be admissible in court and is able to withstand scrutiny.

1. Chain of Custody

I record all the information about the treatment of evidence, and it includes:

- Who collected the evidence
- The date and time of collection
- The actions performed

This makes sure that the evidence can be followed along the process of investigation.

2. Documentation

Detailed records are kept of all observations, tools, and procedures. This encompasses states of the system, commands that were executed and the steps that were taken in the acquisition. Transparency and reproducibility is guaranteed by proper documentation.

3. Hashing

Once I have obtained data such as a memory dump, I compute hash values (MD5 or SHA-256) to verify integrity. These hashes are digital fingerprints, which make certain that the evidence has not been changed in the process of collecting it or even after the collection itself.

4. Repeatability

The whole process must be carried out in a way that will enable a different investigator to repeat the steps and the same results will be obtained thus ensuring reliability and credibility.

PART C: ADAPTIVE EVIDENCE COLLECTION PLAN

Scene Assessment & Reconstructed Timeline

Before formulating the collection strategy, I need to first analyze the situation by reconstituting the chain of events from the dataset. It will make sure that my collection is situation-aware and driven by the concept of evidence volatility and reliability. The following is the sequence of events based on my analysis of the provided dataset:

Table 1. Reconstructed Timeline of Events Based on Dataset Analysis

Time	Event	Source
2025-03-12 13:00	Ahmad Fauzi logged into workstation	system_log.txt
2025-03-12 14:00	Ahmad Fauzi accessed client_database.csv	system_log.txt
2025-03-12 14:32	SanDisk 32GB USB inserted into workstation	usb_log.txt
Unknown	Browser accessed dropbox.com/upload and drive.google.com	history.txt
Unknown	Email sent to external.agent@gmail.com with attachment	suspicious_email.eml
Unknown	client_database_deleted.csv found — deletion attempt detected	Deleted folder

Critical Observation: There is a significant 32-minute time period from when the file was accessed at 14:00 and when the USB was inserted at 14:32. It appears that the suspect used a layered approach in order to exfiltrate data first through the cloud then onto the USB.

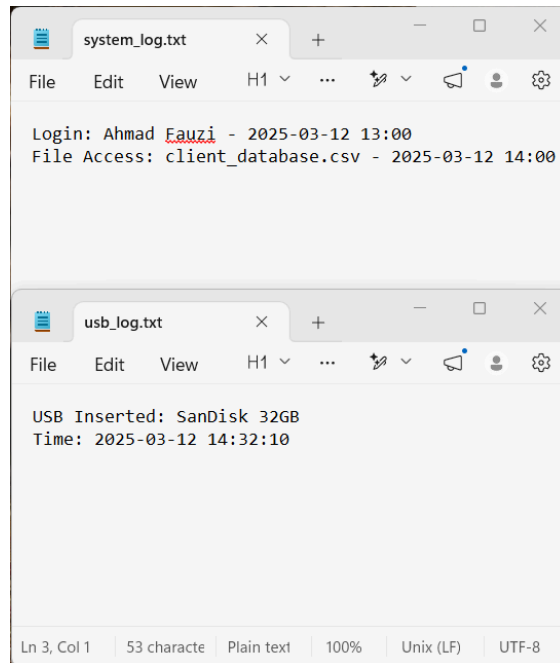


Figure 8. Comparison of system_log.txt and usb_log.txt showing a 32-minute gap between file access and USB insertion

Evidence Collection Priority (What FIRST & Why)

Based on the RFC 3227 Order of Volatility, I will start collecting evidence that is highly volatile since it may quickly become unavailable due to any interference with the crime scene.



Figure 9. RFC 3227 Order of Volatility Diagram — illustrating the hierarchy from most volatile (RAM) to least volatile (archival media)

The table below is a listing of the order in which all the evidence at the scene will be acquired, starting from the most volatile to least volatile.

Table 2. Evidence Prioritization Based on RFC 3227 Order of Volatility

Priority	Evidence Source	Dataset Reference	Justification
1st	Live RAM & Running Processes	(live capture)	Deleted instantly after shutdown; might include session cookies, Dropbox login, and clipboard content
2nd	Live Network Traffic	insider_traffic.pcap	Live exfiltration might still be taking place; can't be

			restored after disconnecting
3rd	Browser History	history.txt	Proves cloud uploading took place; somewhat volatile depending on browser configuration
4th	USB Device Image	usb_log.txt + Physical SanDisk	Connected device remains; possibility of removal by culprit or witnesses
5th	Email Evidence	suspicious_email.eml	Indicates external communications with attachment; shows intent to steal information
6th	System & USB Logs	system_log.txt, usb_log.txt	Determines the chronology of events; rather stable but needs to be secured early
7th	Documents & Deleted Files	client_database.csv, salary_records.csv, client_database_deleted.csv	Data stored on hard disk - most static, yet vital for determining what was stolen.

Adaptive Approach Based on Evidence Type

A) Live System — RAM & Active Processes

Scene Situation:

Based on information gathered from system_log.txt, the session of Ahmad Fauzi remains active. The browser history shows access to the cloud platform and therefore, there could be an ongoing uploading session. It should be noted that in no way will I turn off the computer at this point.

My decision: I have chosen to conduct live acquisition and not to power down the machine.

In case I decided to shut down the computer, I would:

- Permanently lose data stored in RAM – Dropbox/Google Drive session tokens
- Potentially start the execution of any anti-forensic scripts that run at shut-down
- Lose evidence related to open files and clipboard content

Collection Methods:

1. Take a screenshot of the computer screen right away — record the live session graphically
2. Use RAM dump using my forensic USB – never by using the disk of the suspect
3. Take a snapshot of active processes – identify Dropbox.exe or Google Drive sync client process
4. Document all open files – make sure client_database.csv file remains open
5. Document all items in clipboard – may have information copied from database file
6. Record all network connections prior to disconnecting

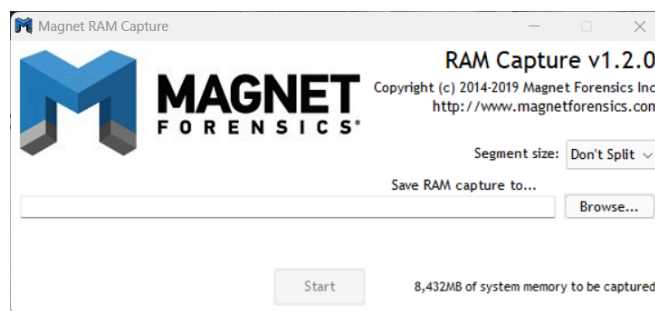


Figure 10. Magnet RAM Capture interface — showing RAM acquisition process running from forensic USB

The following table describes the tools for live system acquisition that I will employ, along with how they relate to this case specifically and what dangers are posed by misuse:

Table 3. Forensic Tools for Live System Acquisition

Tool	How I Would Apply It in This Case	Risk if Misused
Magnet RAM Capture	Dumps physical memory to obtain Dropbox/Google Drive tokens and stored credentials	Extracting data from the HDD of the suspect would erase the exact evidence that I am gathering
WinPmem	Alternate RAM dump utility –	Inaccurate configuration

	outputs .raw file readable by Volatility	could destabilize or crash the system
Volatility Framework	RAM Dump Analysis - look for dropbox.exe, open socket connections, client_database.csv in memory	Misleading process results may be generated due to improper plugin choice
FTK Imager (Portable)	Lists currently opened files and memory contents without installing anything	Only the portable version is to be used; installation on the suspect's computer changes the evidence

B) Network Traffic

Scene Situation:

history.txt proves connections to dropbox.com/upload and drive.google.com.

suspicious_email.eml reveals an email containing attachments sent to external.agent@gmail.com. The insider_traffic.pcap is a snapshot of the captured traffic. In my analysis, the exfiltration through multiple pathways may still be occurring.

My decision: Capture the live traffic and then disconnect the machine from the rest of the network.

In my opinion, it would be wrong to isolate the network before capturing as the live data will get lost during isolation.

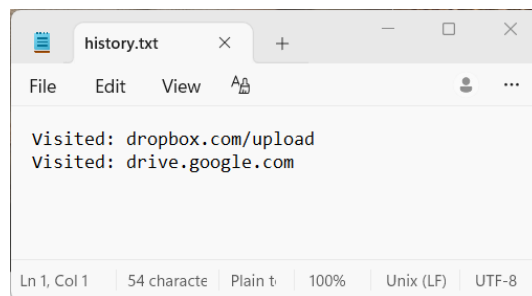


Figure 11. Browser history.txt — showing visits to dropbox.com/upload and drive.google.com as evidence of cloud exfiltration

My collection steps:

1. Connect forensic laptop to network switch using SPAN port via port mirroring — without interrupting existing connection of the suspect
2. Start packet capture right away using either tcpdump or Wireshark
3. Apply filter according to the IP address of the workstation
4. Capture the following:
 - HTTP POST to dropbox.com (uploads)
 - SMTP traffic to gmail.com (e-mail with attachment recreation)
 - DNS query for cloud storage servers
 - Amount of outgoing traffic (to determine how much information has been uploaded)
5. Once I confirm that the packet capture has started – disconnect the workstation from the network
6. Save captured packets in .pcap file and SHA-256 hash

Steps to analyze insider_traffic.pcap: To analyze the above mentioned packet capture file, I will use the following filters in wireshark:

- smtp: To identify email traffic
- http.request.method == POST: To identify POST traffic (file upload to Dropbox)
- dns: To detect external DNS traffic
- ip.dst != [internal range]: To isolate all outbound traffic

Table 4. Forensic Tools for Network Traffic Analysis

Tool	How I Would Apply It in This Case	Risk if Misused
Wireshark	Open insider_traffic.pcap and apply filters for SMTP and HTTP POST protocols to restore any data transfer activity	I must not alter the original file; instead, I would examine a duplicate copy

tcpdump	Real-time sniffing: tcpdump -i eth0 host [suspected IP address] -w capture.pcap	Capture without adequate timestamping reduces its validity in court
NetworkMiner	Restore files that have been transferred through the network – extract email attachments from pcap	I would use Wireshark to confirm results since protocol identification may sometimes fail
Zeek (Bro)	Keep track of any connections and identify any abnormal patterns of traffic	Output should be analyzed carefully as mistakes may be made when reading the logs

Key Observation from history.txt:

It became clear that two distinct cloud storage services were utilized — dropbox.com/upload and drive.google.com. The redundancy here implies multiple exfiltration attempts, which points to the suspect employing different methods to transfer information outside, just in case one method failed.

C) USB Device

Scene Situation:

According to usb_log.txt: "USB Inserted: SanDisk 32GB — Date/Time: 2025-03-12 14:32:10". It is evident that the USB device was inserted 32 minutes after access to client_database.csv. Furthermore, client_database_deleted.csv proves that the suspect tried to delete files after duplicating them, showing that he knew about forensic techniques.

My decision: Conduct an imaging procedure on the USB device using a hardware write blocker rather than forcefully unplugging it.

Forcible disconnection may damage the file system on the USB device along with the deleted files.

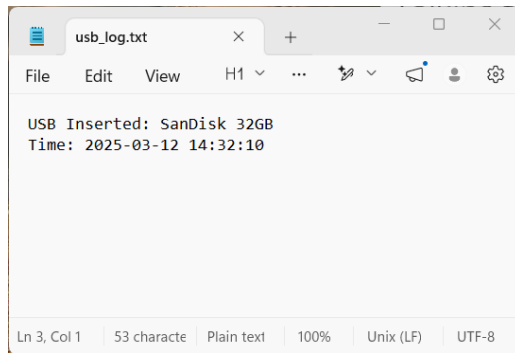


Figure 12. usb_log.txt — showing SanDisk 32GB insertion timestamp at 14:32:10

My Collection Steps:

1. Capture photograph of USB stick while it is still plugged into system — note brand, model, and physical condition before handling
2. Apply my hardware write-blocker (Tableau T8u) prior to accessing drive
3. Capture forensic image of USB stick: `dcfldd if=/dev/sdb of=/evidence/usb_image.dd hash=sha256 hashlog=usb_hash.log`
4. Capture SHA-256 hashes

```
sha256sum /dev/sdb > usb_source.hash
```

```
sha256sum /evidence/usb_image.dd > usb_image.hash
```

5. Confirm that both hashes match — guarantees my image is exact duplicate of evidence
6. Examine forensic image with Autopsy for:
 - Find `client_database.csv` and `salary_records.csv` files
 - Restore deleted files from unallocated space
 - Examine file create/last modified timestamp

The table below summarizes the forensic software I would use to acquire and analyze the USB drive:

Table 5. Forensic Tools for USB Device Acquisition

Tool	How I Would Apply It in This Case	Risk if Misused
Tableau T8u Hardware Write Blocker	Prevent Windows from timestamping the SanDisk or creating any kind of autorun file	Otherwise, plugging in the USB into any system will automatically change the metadata – making my evidence unusable for forensic purposes

dcflddd	Bit-stream imaging with SHA-256 hash at the same time using one command	Replacing if and of flags would destroy my evidence source – an action that is fatal and irreparable
Autopsy	Examines USB image – explores the file system, recovers deleted files, displays file timeline	Analysis should be conducted in the image and not the original USB drive
PhotoRec / Recuva	Deep analysis of unallocated space for recovered copies of client_database.csv	I will run this on the duplicate copy alone since the commands will write the recovery logs on the image

Conclusion Based on Analysis of File: “client_database_deleted.csv”:

Evidence that the file was deleted suggests that the suspect was able to take deliberate action in order to conceal any evidence of their activity after copying the information. I understand that even deleted files are able to be recovered from unallocated disk space before overwriting occurs. It should be recovered for conclusive evidence of the attack.

D) Email Evidence

Scene Situation:

In checking the suspicious_email.eml, I have noticed that it contains the following information:

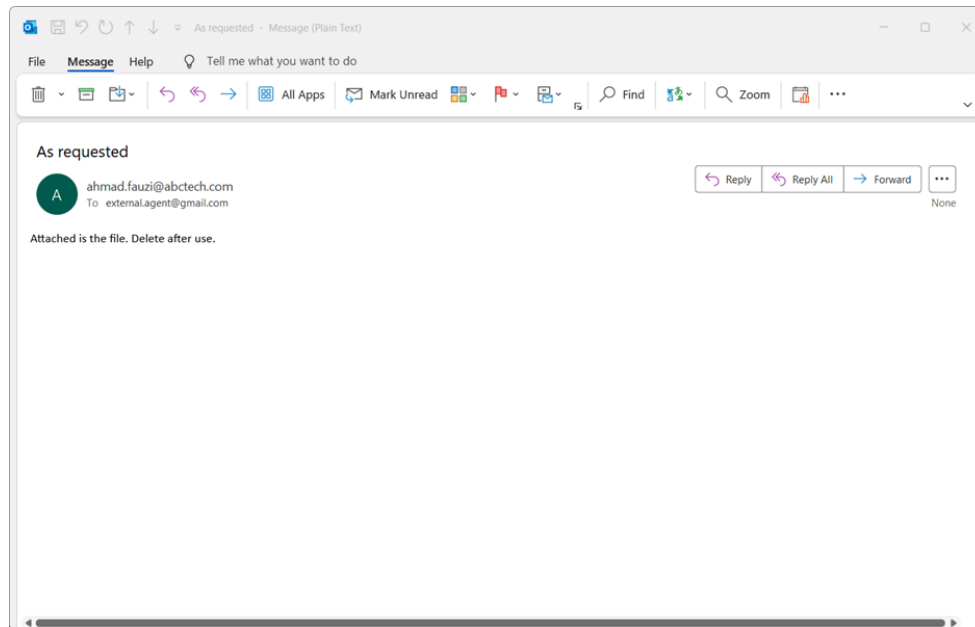


Figure 13. Screenshot of suspicious_email.eml content

This is one of the most important facts I have gathered about this case because "as requested" shows a prior arrangement with someone else, while "delete after use" indicates intent to conceal the criminal activity.

My Collection Steps:

1. Collect the .eml file as it is, without opening it using regular email software
2. Collect email header information:
 - X-Originating-IP: sender's IP address
 - Received: indicates the email's route via mail servers
 - Date: date stamp for correlation with system_log.txt
3. Identify the email attachment – compare with client_database.csv entries
4. Compare with insider_traffic.pcap – identify matching SMTP packets
5. Generate a hash of the .eml file (SHA-256) and retain it in my forensic packaging

The below table provides details on the tools I would employ in the forensic analysis of the email evidence:

Table 6. Forensic Tools for Email Evidence Analysis

Tool	How I Would Apply It
EmailAnalyzer / MXToolbox Header Analyzer	Parse headers of an entire email to identify originating IP address and routing mail servers
Autopsy Email Ingest Module	Perform forensics analysis on the .eml file and obtain any attached files while maintaining metadata integrity
Wireshark	Perform cross-checking on SMTP packets from the pcap to validate email transmission and content of attached files

E) Documents and Deleted Files

Scene Situation: The table below outlines the important documents found in the data set, their information, and their significance to the investigation:

Table 7. Summary of Document and Deleted File Evidence

File	Content	Why I Consider It Significant
client_database.csv	20 client records — Name, Email, Phone	Primary data compromised — Identifies what information was exposed
salary_records.csv	20 employee salary records	Secondary breach — Compromised HR data accessed illegally
client_database_deleted.csv	Marked "DELETED DATA"	Verifies presence of anti-forensics tactic — suspect deleted data after making copies

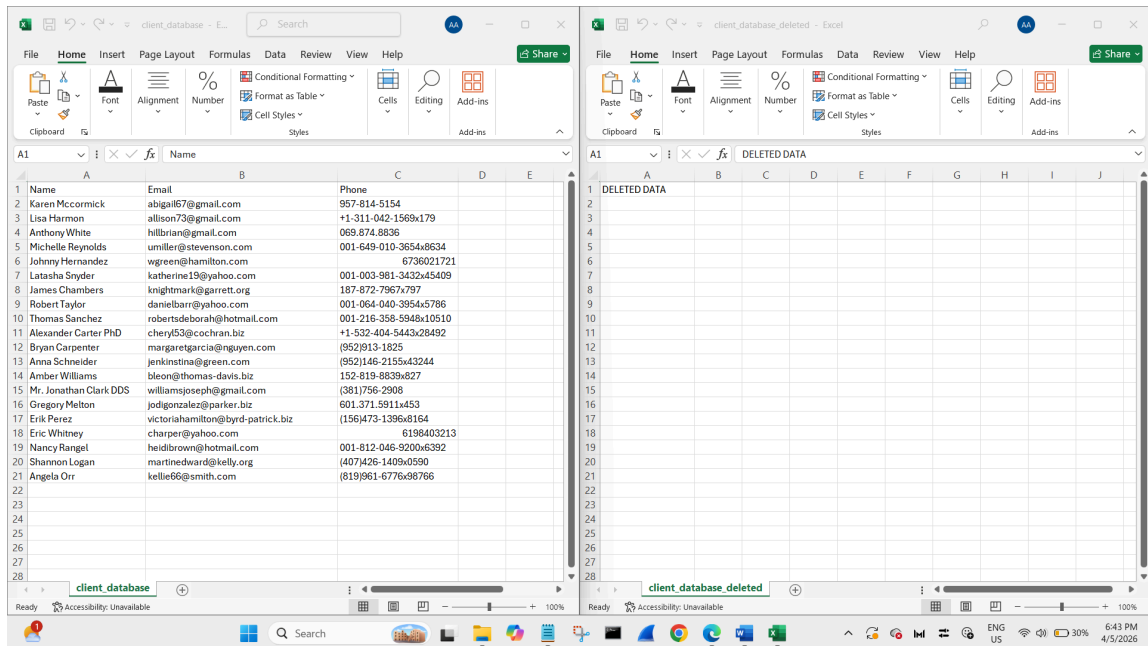


Figure 14. Screenshot of client_database.csv and client_database_deleted.csv – Comparison of original and deleted database records to identify data involved in the breach

My Collection Steps:

1. Perform SHA-256 hash of all files prior to any analysis:

```
sha256sum client_database.csv
```

```
sha256sum salary_records.csv
```

```
sha256sum client_database_deleted.csv
```

2. Make copies for analysis – I will conduct my analysis exclusively using copies
3. Compare client_database.csv with client_database_deleted.csv:
 - Identify deleted records
 - Evaluate whether the deletion was targeted
4. Compare email attachment with database contents: Check if the client_database.csv is the actual file sent to external.agent@gmail.com
5. Gather metadata for all the files:
 - Creation time, modification time, access time (MAC times)
 - Verify MAC times correspond with activities logged in system_log.txt at 14:00

Table 8. SHA-256 Hash Verification Output

File	Hash Value
client_database.csv	d5c48829a767ea154175458fa650c974be9d73e46ddaa3d7718373a27e77221a
salary_records.csv	45af01b547c078f5fce6dd372de2e7a873e3c32a8a7ee2e1ad75e6b2f726ad9c
client_database_deleted.csv	3f935f0dda6ee403b75d295ed6f359ec585471f6904a6287a8feea2840bc8feb

All the evidence gathering techniques employed during the course of this investigation were meticulously planned and performed in adherence to international forensic standards. The order of evidence gathering is determined according to RFC 3227, with all evidence prioritized according to the principle of volatility — ensuring that evidence from more volatile sources, like live RAM and network traffic, were collected prior to moving onto evidence that was relatively less volatile (Bellovin & Killalea, 2002). My actions during the entire evidence-gathering process were also guided by the ACPO Good Practice Guide Principle 1, which requires that every step taken during evidence gathering not in any way affect or tamper with the evidence in question (Association of Chief Police Officers, 2012). Finally, the general principles of digital evidence identification, acquisition, collection, and preservation were informed by the ISO/IEC 27037:2012 standard (ResearchGate, 2025). After analyzing the collected dataset, it can be confidently said that the gathered evidence paints a compelling forensic picture, thus completely proving the scope of Ahmad Fauzi’s premeditated insider data breach.

PART D: LEGAL RISK ASSESSMENT & COMPLIANCE

1. Legal Readiness Evaluation

Prior to conducting digital forensic investigation, the investigator needs to be aware of the legal requirements to obtain valid evidence and to avoid legal issues.

- Authorization (Legal Permission)

The investigator needs to secure a legal entry warrant, a court order or permission from authorities or the organization. This guarantees the legality of the investigation. Failure to do so can lead to evidence being ruled inadmissible in court due to improper procedure.

- Scope of Investigation

This specifies which devices, systems, or data are subject to investigation. This scope should be followed carefully by investigators to avoid legal issues. For instance, if consent is restricted to a laptop, it may be illegal to access external drives, cloud storage, or other users' devices without their consent. Controlling the scope displays professionalism and avoids legal issues.

- Privacy Considerations

Online investigations may require access to private information. It's essential that only necessary data is collected and examined. Unwarranted access to private data could breach data protection regulations and ethics. Adhering to data privacy practices also helps maintain the company's reputation and avoids legal issues.

2. Consequences of Not Following Procedures

Not adhering to legal process and procedures can have costly implications:

- Evidence Inadmissibility in Court

Evidence gathered without permission or beyond the scope of the investigation may be excluded from court. This may undermine the case, despite evidence of malfeasance.

- Breach of Privacy and Data Protection Regulations

Accessing personal or confidential information without permission can result in fines, lawsuits or regulatory enforcement. This impacts the case and the reputation of the forensic investigator and the company.

- Compromised Evidence Integrity

Failure to follow appropriate forensic practices can lead to evidence tampering, damage or contamination. This casts doubt on its validity and admissibility, preventing it from being used in court.

3. Professional Mitigation Strategies

- Ensure Legal Authorization and Documentation

Always secure and check legal authority before initiating the investigation. Document any authorisation paperwork for future reference.

- Operate within the Scope

Understand the scope of the investigation and do not access any systems or data without permission. If necessary, obtain further permission.

- Implement Forensic Best Practices

Follow best practices for forensic investigations, including:

- Chain of custody to track evidence handling
- Forensic imaging rather than direct access to data
- Use of trusted forensic tools

- Access Control and Data Protection

Restrict access to evidence materials. Data should be stored safely and securely.

- Comprehensive Documentation

All steps in the investigation process need to be documented, including tools accessed, time of access and processes followed. This helps to ensure transparency and evidence in court.

Conclusion

Legal considerations are crucial in digital forensic investigations. Obtaining consent, staying within scope and protecting privacy help ensure that evidence is admissible. Through adherence to professional standards and practices, digital forensic investigators can reduce the risk of legal challenges and ensure the integrity of the investigation.

PART E : EVIDENCE INTEGRITY AND PRESERVATION STRATEGY

Chain of Custody

To maintain the integrity of all digital evidence through chain of custody requirements, every transfer of evidentiary material from collection through presentation into the courtroom is documented with the date/time evidence was transferred, by whom transferred, purpose of transfer, condition of evidence at time of transfer, and who is authorized to access the evidence. Any time evidence is accessed will be documented as well. In this case, the main suspect is Ahmad Fauzi due to evidences in System_Log and Suspicious_email:

System_log :

Login: Ahmad Fauzi - 2025-03-12 13:00

File Access: client_database.csv - 2025-03-12 14:00

Suspicious_email :

From: ahmad.fauzi@abctech.com

To: external.agent@gmail.com

Subject: As requested

Attached is the file. Delete after use.

Evidence Documentation Standard

All evidence must be documented in accordance with standard forensic practices including labeling evidence with unique ID, recording the source and method of acquisition, date/time collected, and investigative personnel associated with the evidence. Additionally, screenshots, logs and notes must be maintained as part of the documentation process of the investigation to provide for transparency and reproducibility.

Hashing Strategy

Hashing is used to verify the integrity of digital evidence. During the acquisition of digital evidence a cryptographic hash function (e.g., MD5 or SHA-256) is generated and recorded for each collection of evidence being acquired. The hash value is reconfirmed in

the analysis phase of the investigative process to confirm the integrity of the evidence (such as verifying that the digital evidence has not been altered). SHA-256 is preferred over MD5 due to its increased security; however, MD5 may be utilized for quicker verification when there are no security concerns with using the MD5 hashing method.

Secure Storage Mechanism

To avoid unauthorized changes or access to the digital evidence, a secure environment will need to be established for the storage of the evidence. Evidence must be stored on write-protected media, encrypted media, and require some form of authentication to prevent unauthorized access. Physical storage of evidence should be located in a facility that can be locked, with compliance to access control requirements, while digital storage of the evidence must use secure servers that support audit logging.

If the integrity of evidence is compromised, evidence may be altered, corrupted, or tampered with, leading to inaccurate analytical results. This could lead to **incorrect conclusions** being drawn from the evidence, proving **the case to not be able to be proven**, and/or resulting in potentially having evidence being excluded from presentation in court. Additionally, it could **damage the reputation** of the investigator(s) and the investigative process.

The proposed **preservation method** provides protection from integrity issues through the implementation of **strict chain-of-custody procedures**, **thorough documentation**, and the **utilization of cryptographic hashing** (to allow for the detection of any changes made to the evidence) as a part of preserving digital/evidence. Secure storage mechanisms will provide protection of the evidence from unauthorized access or modification; and through the above operations, the evidence will maintain its authenticity, reliability, and admissibility when utilized within the judicial system.

PART F : PROFESSIONAL CASE RESOLUTION & IMPROVEMENT PLAN

Critical Evaluation of the Investigation Scenario

The scenario depicted above is a plausible insider attack scenario, where an insider accesses sensitive data without permission, transfers data, and attempts to cover up digital footprints. The dataset reveals the suspect (Ahmad Fauzi) accessed confidential data, copied the data to an external USB storage device, and attempted to exfiltrate the data through email and network communication.

However, there are some shortcomings in the incident response and investigation procedures. The involvement of multiple employees interacting with other systems in the vicinity suggests a lack of scene control, which could lead to contamination. Inappropriate engagement with computer systems can modify, overwrite or delete important evidence, resulting in evidence integrity issues.

Further, the system was connected to the network while suspicious activity was underway. This is a significant security oversight as it enables further data theft or system tampering. Failure to isolate the system implies a poor response in real-time.

Moreover, there is no mention that ephemeral evidence such as RAM and network connections was considered. This would lead to loss of critical forensic evidence, making it impossible to fully understand the incident.

Potential Investigator Mistakes

If the right forensic steps aren't taken, the investigation could make a lot of mistakes.

- The system had to be shut down right away, which meant losing important evidence like running processes, memory data, and active network sessions.
- Not separating the system from the network, which lets data continue to be stolen or changed from afar
- Letting people who aren't supposed to be there stay near the system raises the risk of tampering or contaminating evidence.
- Not handling removable media like USB drives correctly, which could change or destroy evidence
- Investigators not writing down enough about what they did, which breaks the chain of custody and could make evidence unusable in court

These errors show how important it is to have structured, controlled, and well-documented forensic procedures.

Weaknesses in Standard Procedures

Traditional digital forensic methods often focus on analyzing data after an incident has happened (static analysis) and don't pay enough attention to the importance of collecting live data. This method doesn't work well in dynamic environments where systems stay on.

Not being able to collect volatile evidence could mean losing:

- There are active network connections that show data is still being sent.
- Running processes that might show bad behavior
- Temporary data kept in memory, such as information that has been decrypted or partially deleted

Also, standard procedures may not stress the importance of making quick decisions, which is very important when working with live systems. Investigators risk losing important evidence in the early stages of the investigation if they aren't flexible.

Improved Professional Investigation Workflow

A more effective and professional workflow is proposed to address these weaknesses :

Step 1 : Secure the Scene

The first step in the investigation is to make the area safe so that no one can get in without permission. All non-essential personnel must leave the area, and access to the affected system must be limited.

This step is very important to keep evidence from being changed by accident or on purpose. Taking pictures and keeping track of the system's status at the scene is important to make sure that the initial condition is preserved for future reference.

Step 2: Capture Volatile Evidence

Data that changes quickly needs to be collected right away because it is very likely to be lost. This includes getting the contents of RAM, finding out what processes are running, and keeping track of active network connections.

It is important to collect volatile evidence because this kind of data is lost forever when the system is turned off. This step gives you a look at what is happening in the system in real time, including possible data exfiltration.

Step 3: Isolate the System

To stop more data from being sent or remote interference, the system should be cut off from the network. The system, on the other hand, needs to stay on so that it can keep getting volatile data.

Network isolation keeps the system's current state safe and stops any further compromise.

Step 4: Acquire Non-Volatile Evidence

After volatile data has been secured, forensic imaging techniques are used to get non-volatile evidence. This means making exact copies of:

- The drive that stores the system
- The USB device that is connected
- Logs, emails, databases, and other files that are important

Use forensic tools like FTK Imager to make sure that the original data stays the same during acquisition.

Step 5: Correlate Evidence

All collected evidence is analyzed and correlated to reconstruct the sequence of events.

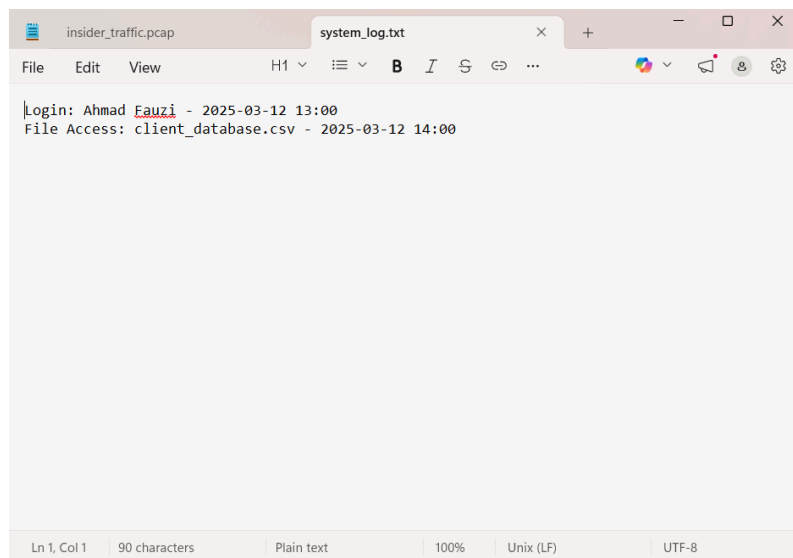


Figure 1 : System log showing database access by Ahmad Fauzi

This figure confirms that the suspect accessed the client database prior to the data exfiltration attempt.

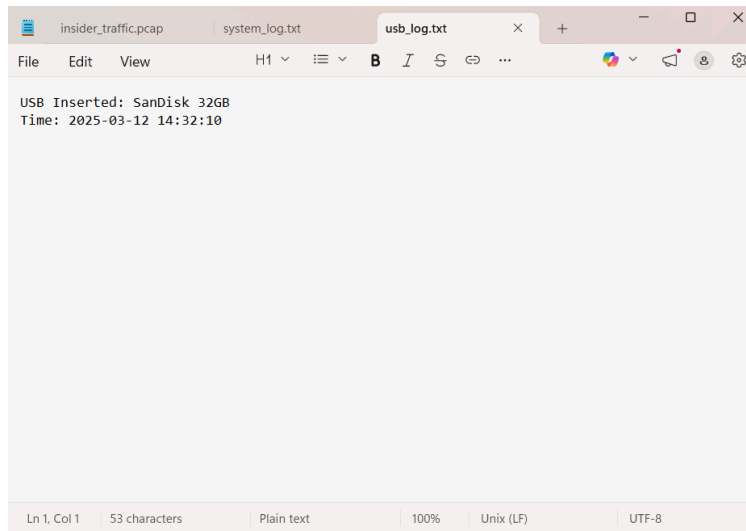


Figure 2 : USB Insertion Log

The USB insertion log confirms that external storage was used immediately after database access, indicating potential data exfiltration

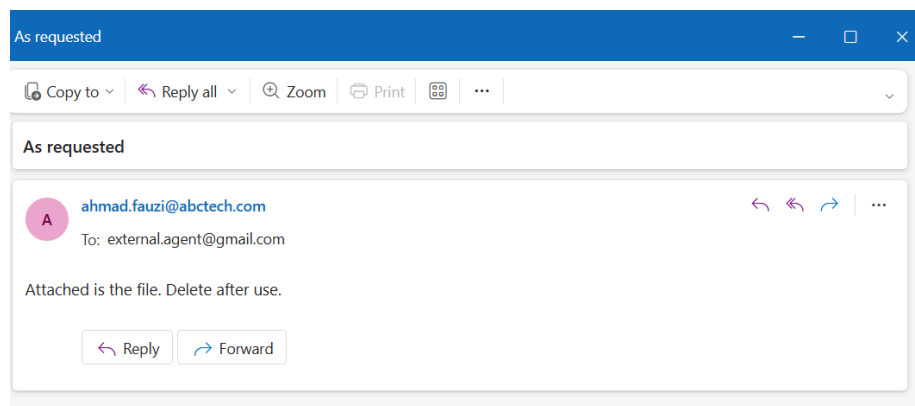


Figure 3 : Suspicious email indicating data exfiltration intent

The email content suggests intentional transfer of confidential data to an external party.

Step 6 : Reporting and Documentation

All findings must be written down in an professional way. This means keeping a record of everything writing down all actions taken and making sure everything is transparent, during the investigation.

Good documentation makes sure that the evidence can be used in court and that the investigation process can be checked and repeated if needed.

Risk Mitigation Strategies

To improve the reliability and integrity of future investigations, several strategies are recommended:

- Enforce strict access control at the crime scene to prevent unauthorized interaction.
- Use write-blocking tools during data acquisition to preserve original evidence.
- Perform live acquisition before system shutdown to capture volatile data.
- Maintain continuous and detailed documentation of all investigative actions.
- Verify evidence integrity using hashing techniques like MD5 or SHA-256.

These measures reduce the risk of evidence loss, contamination, and legal challenges. They ensure that the investigation remains forensically sound.

Professional Justification of Decisions

The decisions in this investigation are based on three key forensic considerations:

- **Evidence preservation:** Volatile data takes precedence to ensure important data is not lost
- **Forensic integrity:** Using tested tools and processes ensures evidence is not changed
- **Legal compliance:** Documentation and procedures ensure legal standards are met

The choice to conduct live acquisition, rather than shut down, is especially warranted in the context of live processes and network connections. This ensures the greatest opportunity to preserve evidence without compromising investigation integrity.

Conclusion

To sum up, the enhanced investigation process presents a well-rounded and professional method of responding to insider data breaches. The approach of gathering volatile evidence, preserving the scene, and employing systematic methods of evidence collection, ensures a reduced risk of data loss and contamination.

The use of real-time decision-making, evidence correlation and documentation showcases a seasoned forensic approach that adheres to industry best practices. This not only enhances the accuracy of the investigation but also assures the admissibility of evidence in court.

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